IT-SIMPLERA INSTITUTE

Network Administration Internship Program

Week 01 Report — Networking Fundamentals & Network Simulation Tools

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Program	Network Administration Internship
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Week	01
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Tools Used	Cisco Packet Tracer 8.2.1, GNS3 2.2.40.1, PuTTY

1. Introduction

This report documents the Week 1 task of the Network Administration Internship at IT-Simplera Institute, which focused on building foundational skills in computer networking and gaining hands-on experience with two industry-standard network simulation platforms: Cisco Packet Tracer and GNS3. The task required designing a simple Local Area Network (LAN) topology, assigning static IP addressing to connected hosts, and verifying end-to-end connectivity using the ping utility.

As advised by the supervisor, the LAN topology was implemented using GNS3's built-in Ethernet switch and VPCS (Virtual PC Simulator) nodes, removing the need to source a separate Cisco IOS image for this particular exercise. The same topology was then replicated in Cisco Packet Tracer to compare the two tools.

2. Networking Concepts Learned

- LAN (Local Area Network): a network confined to a small geographic area, such as a lab or office, connecting multiple hosts through a switch.
- Network devices: a switch operates at Layer 2 and forwards traffic based on MAC addresses, allowing multiple end devices to communicate within the same broadcast domain.
- IP addressing: each host was assigned a unique static IPv4 address within the same subnet (192.168.1.0/24) so they could communicate directly without a router or gateway.
- Subnetting (/24): the /24 prefix indicates a 24-bit subnet mask (255.255.255.0), providing 254 usable host addresses within 192.168.1.0/24.
- Connectivity testing: the ping command uses ICMP Echo Request/Reply messages to verify that two hosts can reach each other across the network.
- Simulation vs. emulation: Packet Tracer simulates network behaviour for learning purposes, while GNS3 can emulate real device operating systems for closer-to-production behaviour.

3. Cisco Packet Tracer — Interface Overview

Cisco Packet Tracer provides a drag-and-drop canvas for placing network devices (switches, routers, PCs, servers) and connecting them with virtual cables. Each end device offers a Desktop tab for IP configuration and a Command Prompt for running diagnostic commands such as ping and ipconfig. The tool also supports a Simulation Mode that visually animates how packets travel across the topology, which is useful for understanding traffic flow step by step.

4. GNS3 — Interface Overview

GNS3 offers a similar canvas-based design area but is built around emulating real device behaviour rather than just simulating it. The left-hand panel provides built-in nodes such as Ethernet switches, Ethernet hubs, and VPCS (lightweight virtual PC nodes) that do not require any external IOS image. Each node can be started, stopped, and accessed through a console (Telnet) session directly from the Topology Summary panel, where commands such as IP assignment and ping tests are executed.

5. GNS3 — LAN Topology and Connectivity Testing

A topology consisting of one Ethernet switch (Switch1) and four VPCS nodes (PC1, PC2, PC3, PC4) was built in GNS3. Each PC was connected to the switch and assigned a static IP address from the 192.168.1.0/24 subnet as follows:

- PC1 — 192.168.1.10/24
- PC2 — 192.168.1.11/24
- PC3 — 192.168.1.12/24
- PC4 — 192.168.1.13/24

5.1 GNS3 Topology

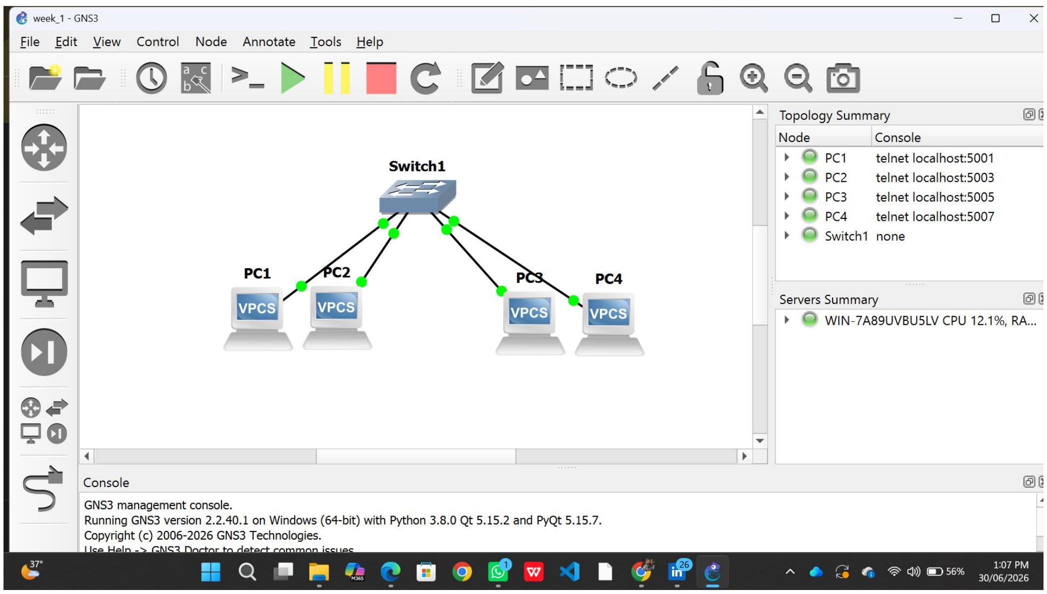

Figure 1: LAN topology in GNS3 — Switch1 connecting PC1 through PC4.

5.2 GNS3 Ping Test Results

From PC1's console, ping tests were run against PC2 (192.168.1.11), PC3 (192.168.1.12), and PC4 (192.168.1.13). All replies were received with 0% packet loss, confirming full Layer 3 connectivity across the switched LAN.

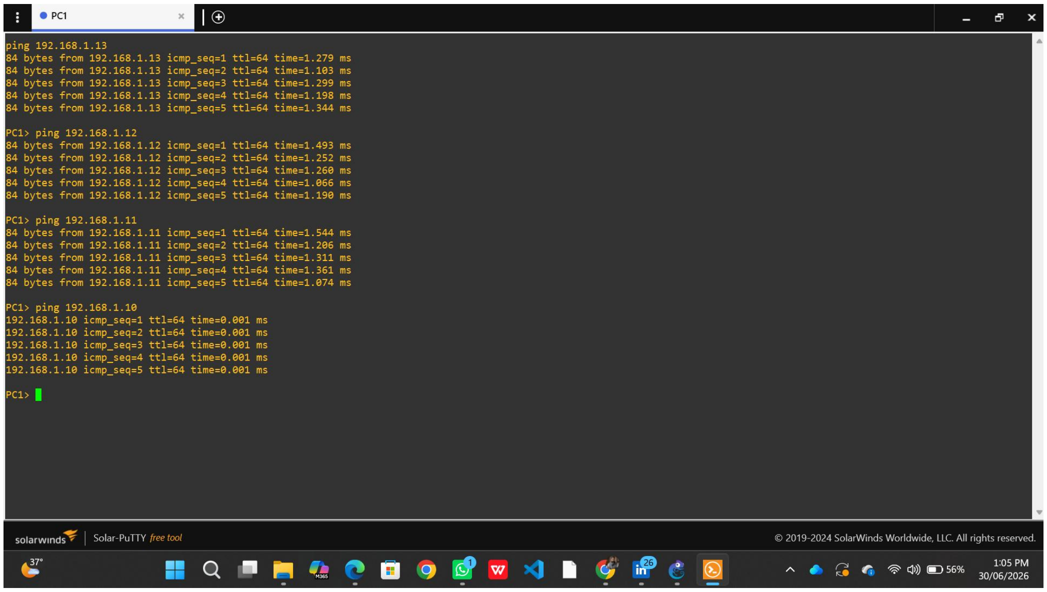

Figure 2: Successful ping results from PC1 to PC2, PC3, and PC4 in GNS3.

6. Cisco Packet Tracer — LAN Topology and Connectivity Testing

The same topology was rebuilt in Cisco Packet Tracer using a Cisco 2960-24TT switch (Switch0) and four PC-PT end devices (PC0–PC3), connected with straight-through copper cables. The same IP addressing scheme (192.168.1.10/24 – 192.168.1.13/24) was applied to keep the comparison consistent with the GNS3 setup.

6.1 Packet Tracer Topology

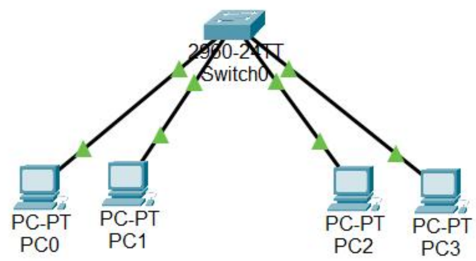

Figure 3: LAN topology in Cisco Packet Tracer — Switch0 connecting PC0 through PC3.

6.2 Packet Tracer Ping Test Results

Using the Command Prompt on PC0's Desktop tab, ping tests were executed against 192.168.1.10, 192.168.1.11, 192.168.1.12, and 192.168.1.13. All requests received replies with 0% packet loss, confirming successful end-to-end connectivity across the topology.

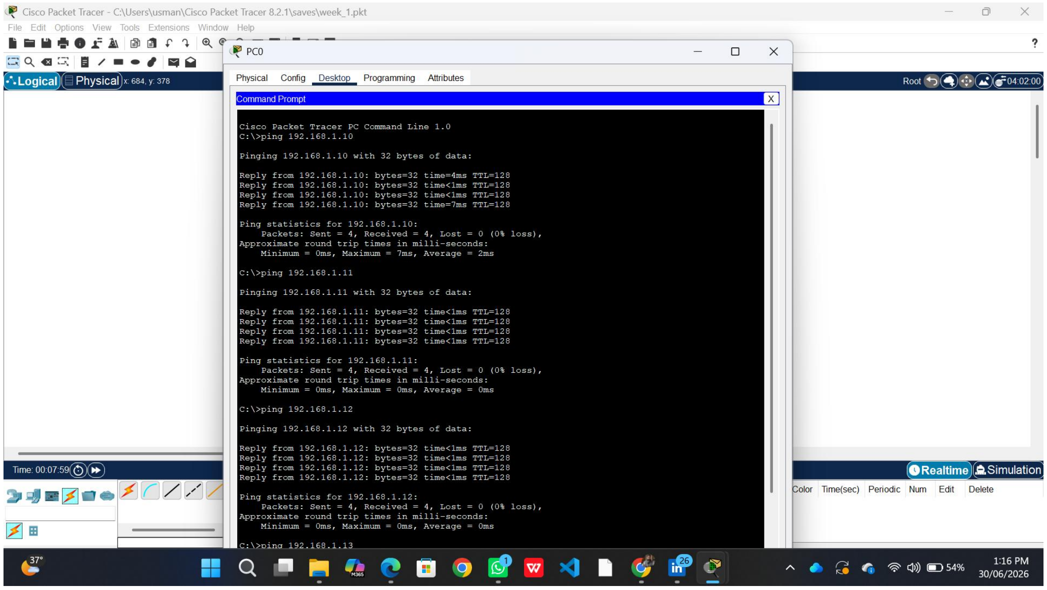

Figure 4: Successful ping results from PC0 to other hosts in Cisco Packet Tracer.

7. Challenges Faced

- Initial uncertainty around GNS3 server modes (local vs. GNS3 VM vs. remote server); resolved by selecting "Run appliances on my local computer" since no Cisco IOS image was required for this task.
- Mistakenly downloaded a VMware ESXi ISO assuming it was needed for the GNS3 VM; clarified that ESXi was unnecessary for this exercise and that VirtualBox is the standard companion for the GNS3 VM when required.
- Encountered a Windows Smart App Control block when extracting a downloaded file via WinRAR; resolved by unblocking the file through its Properties dialog.
- Confirmed with the supervisor that GNS3's built-in switch and VPCS nodes were sufficient for Week 1, avoiding the extra step of sourcing and licensing a Cisco IOS image.

8. Conclusion

This task provided practical, hands-on exposure to two of the most widely used network simulation tools in the industry. Building and testing the same LAN topology in both Cisco Packet Tracer and GNS3 highlighted their core similarities (device placement, IP configuration, and connectivity verification via ping) as well as their differences in depth and realism, with GNS3 oriented toward closer-to-real-world device emulation. Static IP addressing within the 192.168.1.0/24 subnet was successfully configured across four hosts in both platforms, and end-to-end connectivity was verified with 0% packet loss in all ping tests. This exercise has strengthened foundational understanding of LAN design, IP addressing, and connectivity troubleshooting, forming a solid base for more advanced networking tasks in the coming weeks.